Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Natalie Pesarchick	natpes@udel.edu
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Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

With Christmas coming up and me being away from home for the first time, I need some type of way to organize all of my ideas for what I am going to get everyone and how much it will cost me.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

A lot of the current solutions to this problem are for keeping track of your own gift wishes, not other peoples. The best solution I have found is to use any type of spreadsheet, but then I would have to calculate the total cost of everything myself.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

My web application would be a Christmas gift-planning tool that allows people to organize gift ideas for family and friends in one place. Users could create a list for each person, add potential gifts and their prices, track which gifts they have purchased, and automatically

calculate the total cost of all gifts. The application would also allow people to set a budget and see how much they have spent and how much they have left. The goal is to make Christmas shopping easier and more organized, especially now that I am in college and have hometown friends, new friends, and family to be shopping for.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program will ultimately take in a total budget, then repeatedly take in a gift and its corresponding price, add that price to the total to see if the gift is within budget, then if it is, it will add the gift to a checklist where users can check off when they buy the gifts.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

A gift and its corresponding price.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will use the sum pattern to total the cost, then if the total goes over budget, it will tell the user that buying the gift would put them over budget.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back a list of their gifts for each person, the total cost to buy all of them, and how much money they have left in the overall budget.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The main thing that needs to be kept track of are the gifts, people, prices, and budget.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

I need the gift name, gift price, and recipient

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The part that seems the most difficult about this project is putting all the functions together into one code. I am confident I can make them all individually, but working together will be hard.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I am going to try and code everything to the best of my abilities, and if it doesn't run or an error pops up, Im going to have an LLM to help me figure out what went wrong for the sake of saving time. I also am going to have it help me with coding the actual drafter part of the project since that is very new.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

Since the project is due before Christmas time, I want to actually be able to use my program to make my life easier during my Christmas shopping, and many Secret Santas.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to make a website that helps track what you can get your family and friends for christmas and also helps you with budgeting. What are some things that would be helpful to you in a Christmas list organizer for others?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

My friend told me to organize it by person or price range and mark items as bought already so you don't buy more than one. He also said marking the type of gift like clothing , electronics, etc. could be helpful for the user and some things you could reference is spreadsheets

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I think marking the gifts by type would make things more complicated than they already are since it would just be a lot more time consuming. When I asked for a simpler alternative he told me to just keep it grouped by price because the more expensive things (like electronics) would be all put together anyways.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I definitely do like the idea of putting them in price order so if you do include something that is over budget that is a really good gift idea, you can see the most expensive things easily to see if they are even worth buying.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck, I plan on either asking an LLM to help me, or my friends who are also in the same major and taking this class.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

If an LLM provides code that gives me a problem, I am going to ask the LLM to break the code down for me step by step until I can identify the error its telling me. If not, I plan to go to office hours to get help.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

What are the nouns of my problem / what are the core things my program needs to track?

The core things my program needs to keep track of are people, gifts, prices, and the overall budget. Each person can have multiple gifts on their list, and each gift will have a name, a price, and a status showing whether it has been purchased. The program will also keep

track of the total cost of all the gifts and how much money is left in the budget.

If im doing a coding project and AI gives me the wrong code, how do I debug that

If AI gives you code that does not work, you can ask the AI to break down the code step by step and explain what each part does and where the error might be. You can read the error message, check my code for mistakes, and test small parts of the program individually to find the problem. Comparing the code with my class notes to make sure it follows what you have learned is also a great idea! Once you understand the mistake, you can fix the code and test it again to make sure it works correctly.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.